

Automated Retinal Blood Vessel Segmentation Using U-Net

Complete Technical Review & Implementation Baseline (DRIVE Dataset)

Based on Ronneberger et al. (2015)

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1. Problem Statement & Clinical Motivation

Clinical Significance

- Retinal vascular structures provide non-invasive biomarkers for CAD:
 - ▶ **Diabetic Retinopathy:** Neovascularization & microaneurysms
 - ▶ **Glaucoma:** Disc-to-vessel ratio changes
 - ▶ **Hypertension:** Arteriovenous nicking & vessel narrowing
 - ▶ **AMD:** Vascular anomalies near the macula

Proposed Solution

- Implementation of original **U-Net (Ronneberger et al., 2015)**.
- End-to-end fully convolutional network (FCN).
- Handles sparse data via elastic deformations & patch sampling.
- Full-resolution prediction via **Overlap-Tile Strategy**.

Core Segmentation Challenges

- **Severe Class Imbalance:** Vessels cover only 10–15% of FOV pixels.
- **Complex Topography:** Thin, tortuous capillaries (≤ 1 pixel width).
- **Noise & Artifacts:** Variable illumination, optic disc

2. Dataset Specification & Data Leakage Prevention

DRIVE Benchmark Specifications

- **Total Fundus Images:** 40 images (565×584 RGB .tif).
- **Camera FOV:** 45° circular field of view.
- **Official Benchmark Split:**
 - ▶ 20 Training images (with expert ground truth).
 - ▶ 20 Test images (held-out, FOV mask only).
- **Circular FOV Mask:** Crucial for restricting evaluation to valid retinal regions.

Image-Level Split vs. Data Leakage

- **Train/Val Partitioning:** Split at **image-level** (16 Train / 4 Val).
- **Why NOT patch-level split?**
 - ▶ Patches from the same image share structural & illumination correlation.
 - ▶ Patch-level splitting leads to **Data Leakage** → artificially inflated metrics.
- **Guarantee:** Val set images (37–40) are completely unseen during training.

3. Classical Image Preprocessing Pipeline

Pipeline Flow: RGB Input $\longrightarrow$ Green Channel $\longrightarrow$ CLAHE $\longrightarrow$ Bilateral Filter $\longrightarrow$ Normalize [0,1]

1. Green Channel

- Hemoglobin has peak absorption at **540–575 nm**.
- Provides maximum vessel-to-background contrast.
- Red/Blue channels carry excess noise.

2. CLAHE Contrast Boost

- **Clip Limit = 2.0:** Prevents noise over-amplification.
- **Tile Grid = 8×8 :** Local contrast enhancement for fine capillaries.

3. Bilateral Filter

- **Edge-Preserving:** Denoises background while keeping vessel boundaries sharp.
- Parameters: $d = 5$, $\sigma_{\text{color}} = 75$, $\sigma_{\text{space}} = 75$.

4. Data Augmentation & Elastic Deformation

Standard Label-Preserving Augmentations

- **Flips:** Random Horizontal & Vertical ($p = 0.5$).
- **Rotations:** 90° , 180° , 270° random rotations.
- Exposes network to invariant vascular orientation patterns.

Elastic Deformation (Ronneberger et al.)

- Simulates tissue deformation in biological specimens.
- Smooth random displacement grid:

$$\mu = 30.0, \sigma = 4.0$$

Spatial Mismatch Handling

- Input patch is 284×284 ; target mask is 100×100 .
- **Center-Slicing Displacement Field:**

$$\text{margin} = \frac{284 - 100}{2} = 92 \text{ px}$$

- Center region of (dx, dy) is sliced to deform the target mask, guaranteeing perfect alignment with input patch.

5. Patch Extraction Strategy & Valid Conv Math

Mathematical Geometry of Valid Convolutions

- Valid convolution (padding = 0) eliminates boundary artifacts.
- Each 3×3 Conv reduces spatial dimension by 2 px.
- **DoubleConv Block** reduces dimension by 4 px.
- **Network Shrinkage Breakdown:**
 - ▶ 5 Encoder/Bottleneck blocks $\times 4$ px = 20 px
 - ▶ 4 Decoder blocks $\times 4$ px = 16 px
 - ▶ Total reduction = 36 px per side (184 px diameter)
- **Receptive Field Relation:**

$$\text{Output Size} = \text{Input Size} - 184 \implies 284 \rightarrow 100$$

FOV-Guided Patch Sampling

- Mirror padding full image by margin = 92 px using `reflect` mode.
- Center coordinates sampled randomly **strictly within valid FOV**.
- Concentrates training on clinically relevant tissue instead of black background.
- **Patch Count:** 10 patches/img $\times$ 16 imgs = 160 patches/epoch.

6. U-Net Architecture & Design Enhancements

Symmetric Encoder-Decoder Structure

- **Contracting Path (Encoder):** $4 \times$ [DoubleConv + MaxPool2d(2,2)]. Channels: $1 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$.
- **Bottleneck:** 1024 channels at 10×10 resolution + **Dropout(0.5)**.
- **Expansive Path (Decoder):** $4 \times$ [ConvTranspose2d(2,2) + Crop & Cat + DoubleConv].
- **Skip Connections:** Center-crop encoder tensor to resolve shape mismatch from valid convolutions:

$$\text{Crop Margin} = \frac{H_{\text{enc}} - H_{\text{dec}}}{2}$$

Architecture Modifications

- **Batch Normalization:** Added after each Conv2d layer (enhances CPU stability & convergence).
- **Dropout (0.5):** Active at bottleneck during training to prevent overfitting.
- **Output Layer:** 1×1 Conv + Sigmoid activation.
- **Parameters:** 31,036,481 (31.0M).

7. Loss Functions: Addressing Class Imbalance

Soft Dice Loss

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i y_i + 1}{\sum_i p_i + \sum_i y_i + 1}$$

- Directly optimizes spatial overlap.
- Immune to class imbalance — penalizes zero-vessel trivial predictions.
- Smooth Laplace term (+1) prevents division by zero.

Binary Cross Entropy (BCE) Loss

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

- Provides stable per-pixel gradient signals across all regions.

Combined Composite Loss

$$\mathcal{L}_{\text{Total}} = 0.5 \cdot \mathcal{L}_{\text{BCE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}}$$

Combines pixel-wise convergence of BCE with overlap optimization of Dice.

8. Training Engine & Optimization Setup

Hyperparameters & Optimizer

- **Optimizer:** Adam ($lr = 10^{-3}$, weight decay $= 10^{-5}$).
- **LR Scheduler:** `CosineAnnealingLR` ($T_{\max} = E$). Smooth decay to near-zero avoids local oscillation.
- **Batch Size:** 4 patches.
- **Epochs:** 5 (baseline execution) / 50 (full convergence).
- **Training Time:** ~ 13.24 minutes on CPU (5 epochs).

Model Checkpointing Strategy

- Monitored Metric: **Validation Dice Score**.
- Saves state dictionary **only when validation Dice improves**.
- Prevents saving overfitted late-epoch checkpoints.
- Preserves full optimizer & scheduler states for seamless resumption.

9. Full-Image Inference: Overlap-Tile Strategy

Problem: DRIVE images (565×584) exceed input patch dimensions. Resizing destroys fine vessel details.

Seamless Tiling Workflow

- 1 **Mirror Padding:** Extend full image borders by margin (92 px) using `reflect` mode.
- 2 **Tile Sliding:** Slide input window by output stride (100 px).
- 3 **Context Extraction:** 284×284 input tile feeds model $\rightarrow$ outputs seamless 100×100 prediction.

Stitching & FOV Masking

- 1 **Stitching:** Place 100×100 tiles non-overlappingly into full output grid.
- 2 **Crop & Mask:** Crop to original 565×584 dimensions and multiply by binary FOV mask:

$$P_{\text{final}} = P_{\text{stitched}} \odot M_{\text{FOV}}$$

10. Evaluation Metrics & Benchmark Results

FOV-Restricted Metric Computation

- Evaluated **strictly inside circular FOV** ($M_{\text{FOV}} > 0.5$).
- Prevents background pixels from artificially boosting Accuracy & Specificity.
- **Threshold:** $p \geq 0.5 \implies$ Binary vessel.

Validation Benchmark Results (4 Held-Out Images)

Metric	37_t	38_t	39_t	40_t	Mean
Accuracy	0.934	0.942	0.938	0.946	0.940
Sensitivity	0.793	0.736	0.760	0.815	0.776
Specificity	0.955	0.971	0.964	0.962	0.963
Dice (F1)	0.753	0.761	0.754	0.767	0.759
IoU	0.604	0.614	0.605	0.622	0.611
AUC-ROC	0.947	0.959	0.955	0.964	0.957

11. SOTA Comparison & Code Structure

Comparative Performance Baseline

Method	AUC	Sens	Spec	Acc
Staal et al. (2004)	0.952	0.718	0.977	0.94
Soares et al. (2006)	0.961	0.733	0.978	0.94
U-Net (Full GPU)	0.979	0.770	0.981	0.95
Ours (5-ep CPU)	0.957	0.776	0.963	0.94

- High sensitivity (0.776) outperforming classical baselines even with short CPU training.

Modular Project Folder Structure

- `src/preprocessing.py`: CLAHE + Bilateral
- `src/dataset.py`: Patch generator & splits
- `src/unet_model.py`: PyTorch U-Net
- `src/losses.py`: Combined BCE+Dice
- `src/metrics.py`: FOV metric calculator
- `src/utils.py`: Overlap-tile inference engine
- `scripts/train.py` & `evaluate.py`: Workflows

12. Summary of Design Decisions & Key Takeaways

Key Engineering Takeaways

- 1 **Data Leakage Prevention:** Image-level dataset splitting is strictly required when evaluating patch-based segmentation networks.
- 2 **Valid Convolutions:** Input dimensions (284×284) must compensate for valid conv edge reduction (184 px) to yield centered predictions (100×100).
- 3 **Loss Function Synergy:** Composite BCE + Soft Dice overcomes severe 85 : 15 background-to-vessel class imbalance.
- 4 **Overlap-Tile Inference:** Mirror padding eliminates tile edge discontinuity, enabling seamless full-resolution prediction.

Thank You! Questions & Discussion